

HCMUT DETAILED PLAN PROPOSAL

AI Model Training for AD L4 Robotaxi Systems

Quantitative Deliverables & Measurable Acceptance Criteria

Client: VinFast - L4 Team

Provider: Ho Chi Minh City University of Technology (HCMUT)

Project Duration: 10 months (September 1, 2026 - June 30, 2027)

Version: 1.0 - Detailed Specification

This proposal provides concrete, measurable deliverables and acceptance criteria for all project milestones to ensure transparent evaluation and successful deployment.

PROJECT OVERVIEW

In order to foster effective collaboration and promote technological advancement in the development of Level 4 (L4) Autonomous Driving (AD) systems, Ho Chi Minh City University of Technology (HCMUT), in partnership with VinFast's L4 Team, will undertake a research and development project focused on developing a World Model architecture for autonomous driving systems.

This project aims to transition from computationally intensive 3D external simulators toward an evaluator-free latent closed-loop simulation framework, optimized for dense and complex traffic environments in Vietnam, particularly those characterized by a high density of motorcycles.

Project Structure

The project is structured into **three main phases**:

- **Phase 1** (3 months): Development of CaMo-JEPA for Self-Supervised Perception and Confounder Estimation.
- **Phase 2** (4 months): Action-Conditioned Latent SCM Predictor & Anytime-Valid Causal Planner
- **Phase 3** (3 months): Evaluator-Free Closed-Loop Evaluation, Fine-Tuning & Deployment

The project commences at HCMUT on **September 1st, 2026**, with periodic deliverables provided to VinFast at each milestone to ensure alignment and continuous progress towards system integration and deployment.

Key Differentiator: Quantitative & Measurable Deliverables

This proposal differs from typical proposals by providing:

- 1. **Specific performance metrics** with numerical thresholds (e.g., GPU FLOPs reduction $\geq 60\%$, Latent Drift ≈ 0)
- 2. **Concrete output formats** (Latent Vectors, trajectories, MP4 rollout videos)
- 3. **Quantified test coverage** (minimum driving hours, minimum rollout trajectories, minimum e-testing trials, dense scenarios)
- 4. **Objective acceptance criteria aligned with Sequential e-Testing** (minimum driving hours, minimum rollout trajectories, minimum e-testing trials)
- 5. **Measurable resource-efficiency KPIs**, since GPU cost reduction is a first-class deliverable of this architecture, not a side effect

This ensures **transparent evaluation, clear acceptance gates, and successful deployment.**

PHASE 1: CAMO-JEPA PERCEPTION (3 MONTHS)

Team Composition

To effectively deliver the objectives in Phase 1, the HCMUT team will consist of:

Phase 1 staffing and cost

Role	Headcount	Monthly Cost (VNĐ)	Total (3 months) (VNĐ)
Project Leader	1	13,500,000	40,500,000
Teaching Assistant	2	12,000,000	72,000,000
Student	5	6,000,000	90,000,000
Total	8		202,500,000

Responsibilities:

- **Project Leader:** Overall coordination, architecture design, causal-inference guidance, reporting

- **Teaching Assistants:** Data pipeline engineering, model training, factorization/confounder evaluation, documentation
- **Students:** Data collection support, experiments, evaluation

Milestone 1: Proposed CaMo-JEPA Perception & Confounder Pipeline

Deadline: November 30th, 2026

Input Data

- Multi-sensor data from VinFast VF8 sensor suite:
 - 8 multi-view cameras (RGB)
 - 12 LiDAR sensors
 - Radar, IMU, GNSS data
- VinFast proprietary dataset: Minimum **100 hours** of continuous driving data collected on dense residential and mixed-traffic routes, including narrow streets, motorcycle-heavy intersections, and market streets. The dataset is expected to comprise 70% training data and 30% test data, with coverage targets of 50% clear daytime traffic conditions, 25% adverse weather conditions, and 25% high-traffic-density scenarios.

Quantitative Performance Metrics

World Model performance requirements

Metric	Target	Evaluation Method
Old Knowledge Retention	$L_{recon} \approx 0$	Cosine similarity ($L_{recon} = 1 - \cos(z_t, e_{static})$) against original ViT
Static-scene mIoU retention (lane, divider, sky) vs. original ViT baseline	$\geq 95\%$ mIoU of original	Linear-probe segmentation head, frozen backbone
Near-body motorcycle detection (Linear Probe),	mAccuracy $\geq 80\%$	500+ manually spot-checked near-body motorcycle events

Metric	Target	Evaluation Method
mAccuracy, Flow branch ON		
Near-body motorcycle detection improvement, Flow ON vs. Flow OFF (IoU)	$\geq +15$ pts of IoU	Flow ON vs. Flow OFF ablation
Latent Orthogonality	$L_{orth} \approx 0$	Measure Difference Loss $L_{orth} = \left\ z_{task}^T \cdot z_{exo} \right\ _F^2$
Mutual Information, $I(z_{exo}; a_t)$	$I(z_{exo}; a_t) \leq 0.05$ nats	InfoNCE/MINE estimator, 3 independent runs: Prove independence of exogenous variables from ego-action
GPU backprop cost reduction vs. full ViT fine-tuning	Reduce by 70%	TFLOPs measured during training

Inference Performance Requirements

Perception backbone inference requirements

Component	Platform	Max Latency
Frozen ViT (16×16) encoder, 8 cams	RTX 4090	≤ 60 ms (batched)
FlowFormer++ flow branch, 8 cams	RTX 4090	≤ 30 ms (batched)
Fusion + Causal Factorization	RTX 4090	≤ 10 ms
Total perception latency	RTX 4090	≤ 100ms

Component	Platform	Max Latency
Total perception latency	Jetson AGX Orin	$\leq 250\text{ms}$
Minimum FPS		≥ 10 FPS

Model Interface (Internal Phase 1 to Phase 2 Data Flow)

All outputs must be in **standardized, machine-readable formats**:

1. Latent Tensors:

- Format: PyTorch Tensor (torch.Tensor) $z_t \in \mathbb{R}^{(N \times D)}$, decomposed and saved separately as $z_{task,t}, z_{exo,t}$
- Sampling: Temporally aligned to the sliding window FrameBatch clips (default $T = 16$ frames, stride = 1)

2. Confounder Vector:

- PyTorch Tensor (torch.Tensor) $U_t \in \mathbb{R}^{(D_U)}$ with $D_U = 128$, from GRU Confounder Encoder.
- Exported per-batch as the final hidden state summarizing the $(T - 1)$ temporal history window:

$$U_t = h_{T-1} = GRU(SpatialMeanPool(z_{fused}))$$

Robustness Requirements

Robustness validation requirements

Condition	Min Test Frames	Latent-Drift Target
Clear weather (baseline)	500	≤ 0.05 (reference)
Light rain	500	≤ 0.07
Heavy rain	500	≤ 0.08
Fog	500	≤ 0.10
Night (low light)	500	≤ 0.10

Condition	Min Test Frames	Latent-Drift Target
TOTAL	2,500+ frames	

Maximum accuracy drop: $\leq 15\%$ relative increase in Latent Drift across all adverse conditions vs. clear-daylight baseline

Phase 1 Deliverables (Specific Formats)

1. Diagnostic Visualization:

- Optical-flow heatmap overlay video (MP4) highlighting near-body motorcycle motion, minimum 5 demo clips, 30–60s each
- Cosine-similarity drift plot per driving segment (PNG/CSV)

2. Model Checkpoints:

- PyTorch:
 - camo.pt: The most recent checkpoint (Overwrite at the end of each epoch).
 - best_camo.pt: The checkpoint with the best (lowest) Average Loss to date.
 - step_{batch_idx}.pt: Mid-epoch checkpoint saved after a certain number of steps.
- State Dict: Strictly contains trainable modules only (flow_encoder, fusion, factorizer, confounder, predictor) to minimize file size; frozen backbones are excluded.
- ONNX export (optional): camo.onnx

3. Training Scripts:

- Reproducible training: fixed random seed, config YAML/CaMoJEPACONFIG with all hyperparameters (freeze flags, loss weights, ablation flags, ViT & FlowFormer++ paths)
- Training logs: TensorBoard format, ≥ 50 epochs
- Result variance: $\leq 2\%$ across 3 independent runs

4. Evaluation Scripts:

- Cosine-similarity / Latent-Drift calculator (CLI, JSON/CSV output)

- Orthogonality-loss and Mutual-Information (or action-leakage proxy) estimator scripts
- Ablation harness: motion branch off, confounder off ($U = 0$), factorizer off
- Linear-probe evaluation harness for near-body / small-motion (e.g. motorcycle), comparing static-only vs. static+motion
- Resource script: GPU memory and time/step under frozen ViT + frozen FlowFormer++

5. Unit Tests:

- Test coverage: $\geq 80\%$ for causal modules, fusion, losses, and phase1 wiring
- Framework: pytest or unittest
- Automated CI: GitHub Actions or equivalent

Dataset Requirements

Perception-phase data coverage

Source	Volume	Purpose
VinFast 8-cam footage (current)	20 minutes	Initial validation, sanity checks
VinFast new 8-cam collection	≥ 100 hours	Including both training and test sets, sufficient for statistically meaningful probing and MI estimation
Legacy NAVSIM dataset	Reused, existing set	Auxiliary comparison only (not required for training)
Total new raw video	≥ 100 hours + legacy	Comprehensive validation

PHASE 2: ACTION-CONDITIONED PREDICTION & CAUSAL PLANNING (4 MONTHS)

Team Composition

For Phase 2, HCMUT will expand the team to cover predictive modeling and reinforcement-learning planning workloads:

Phase 2 staffing and cost

Role	Headcount	Monthly Cost (VNĐ)	Total (4 months) (VNĐ)
Project Leader	1	13,500,000	54,000,000
Teaching Assistant	3	12,000,000	144,000,000
Student	5	6,000,000	120,000,000
Total	9		318,000,000

Deadline: March 31st, 2027

Milestone 2.1: Action-Conditioned Latent SCM Predictor

Input Data

- $z_{task,t}$, $z_{exo,t}$, U_t sequences from Milestone 1 output, aligned with ego-action $\log a_t$ (steering, throttle/brake)
- Multi-sensor data from VinFast VF8 sensor suite:
 - 8 multi-view cameras (RGB)
 - 12 LiDAR sensors
 - Radar, IMU, GNSS data
- VinFast proprietary dataset: Minimum **100 hours** of continuous driving data collected on dense residential and mixed-traffic routes, including narrow streets, motorcycle-heavy intersections, and market streets. The dataset is expected to comprise 70% training data and 30% test data, with coverage targets of 50% clear daytime traffic conditions, 25% adverse weather conditions, and 25% high-traffic-density scenarios.

Quantitative Performance Metrics

Latent SCM predictor performance requirements

Metric	Target
1-step latent prediction error (L2, normalized)	≤ 0.08
Causal Dynamics Loss (vs. ground-truth z_{t+1})	≤ 0.10
3-second rollout ADE (decoded ego-trajectory)	$\leq 1.0\text{m}$
3-second rollout FDE (decoded ego-trajectory)	$\leq 2.0\text{m}$
do(a_t) intervention consistency: $I(z_{\text{exo}} ; a_t)$ post-predictor	≤ 0.05 nats
Redundant rollout-compute reduction (z_{exo} computed once for 32 branches)	30% – 40%

Inference Performance

- **Predictor latency (1 step, RTX 4090):** $\leq 5\text{ms}$ per branch, batched across 32 branches $\leq 30\text{ms}$ total
- **Predictor latency (Jetson AGX Orin, 1 step):** $\leq 15\text{ms}$ per branch, batched across 32 branches $\leq 60\text{ms}$ total

Phase 2.1 Deliverables

1. **Pre-trained Models:**
 - PyTorch checkpoint: scm_predictor_task.pth, scm_predictor_exo.pth
 - ONNX export
2. **Training Artifacts:**
 - Training logs: TensorBoard with ≥ 30 epochs
 - Config files: YAML with all hyperparameters documented
 - Checkpoints: Best and last epoch models
3. **Evaluation Scripts:**
 - ADE/FDE calculator

- Causal Dynamics Loss logger
- JSON metric export

4. Visualization:

- Minimum 5 examples of predicted-vs-ground-truth latent-decoded trajectory overlays

Milestone 2.2: Anytime-Valid Causal Planner (Latent RL Planner)

Input Data

- $z_{task,t}$, $z_{exo,t}$, U_t sequences from Milestone 1 output, aligned with ego-action $\log a_t$ (steering, throttle/brake)
- Multi-sensor data from VinFast VF8 sensor suite:
 - 8 multi-view cameras (RGB)
 - 12 LiDAR sensors
 - Radar, IMU, GNSS data
- VinFast proprietary dataset: Minimum **100 hours** of continuous driving data collected on dense residential and mixed-traffic routes, including narrow streets, motorcycle-heavy intersections, and market streets. The dataset is expected to comprise 70% training data and 30% test data, with coverage targets of 50% clear daytime traffic conditions, 25% adverse weather conditions, and 25% high-traffic-density scenarios.

Planning & Safety Performance

Anytime-valid planner requirements

Metric	Target
Rollout trajectory candidates per planning cycle	32
Rollout Collision Rate (simulated, latent space)	$\leq 5\%$
Off-road / lane-violation rate (simulated)	$\leq 5\%$
Planning latency per cycle (32 branches, RTX 4090)	$\leq 80\text{ms}$

Anytime-Valid Early-Stopping (Ville's Inequality), reduction in rollout steps vs. fixed-horizon rollout	50%–70%
False early-stop rate (safe trajectory wrongly cut)	$\leq 5\%$
Threshold: e-process $E_\tau \geq 1/\alpha$, with α	0.05

Output Format Specifications

1. Trajectory Output:

- Format: JSON, minimum 8 waypoints per trajectory, sampling interval 0.5s
- Schema: `{"trajectory": [{"t":..., "x":..., "y":..., "heading":..., "v":...}], "risk_score": 0.0-1.0, "branch_id": ...}`
- Coordinates in vehicle reference frame

2. Rollout Video:

- Format: MP4, one clip per demo scenario, 1–2 minutes each, minimum **10 demo clips**
- Content: side-by-side predicted latent-decoded rollout vs. ground-truth future frames, with overlaid planned trajectory and risk score

3. Confidence/Risk Score: e-value per trajectory branch, range $[0, 1/\alpha]$, logged per rollout

Phase 2.2 Deliverables

1. **Planner Components:** RL Policy Agent checkpoint, e-process early-stopping module, cost-function weights (YAML, ≤ 15 tunable parameters)
2. **Baseline Comparison Scripts:** automated comparison vs. Drive-JEPA + NAVSIM closed-loop baseline on the reused legacy labeled set
3. **Docker Container:** reproducible environment, tested on clean machine

Dataset Requirements

- Cumulative training data: ≥ 100 hours (8-cam + CAN Bus)
- Legacy labeled (“NAVSIM-style”) set: reused for baseline comparison scenarios, minimum 30 routes

Documentation

- **Technical report:** minimum 20 pages
- **Model architecture:** planner architecture (SCM Predictor + RL Policy + Early-Stopping), pseudocode for Ville's-inequality stopping rule
- **Evaluation results:** tables, precision-recall curves, confusion matrices
- **API documentation:** all functions documented
- **Comparison study:** vs. minimum 2 baseline methods

PHASE 3: EVALUATOR-FREE CLOSED-LOOP EVALUATION, FINE-TUNING & DEPLOYMENT (3 MONTHS)

Team Composition

To effectively deliver the objectives in Phase 3, the HCMUT team will consist of:

Phase 3 staffing and cost

Role	Headcount	Monthly Cost (VND)	Total (3 months) (VND)
Project Leader	1	13,500,000	40,500,000
Teaching Assistant	2	12,000,000	72,000,000
Student	5	6,000,000	90,000,000
Total	8		202,500,000

Responsibilities:

- **Project Leader:** Overall coordination, architecture design, causal-inference guidance, reporting
- **Teaching Assistants:** Data pipeline engineering, model training, factorization/confounder evaluation, documentation
- **Students:** Data collection support, experiments, evaluation

Milestone 3: Latent Closed-Loop Simulation & Sequential e-Testing

Deadline: May 31st, 2027

Input Data

- Full CaMo-JEPA pipeline and output from phase 2
- Multi-sensor data from VinFast VF8 sensor suite:
 - 8 multi-view cameras (RGB)
 - 12 LiDAR sensors
 - Radar, IMU, GNSS data
- VinFast proprietary dataset: Minimum **100 hours** of continuous driving data collected on dense residential and mixed-traffic routes, including narrow streets, motorcycle-heavy intersections, and market streets. The dataset is expected to comprise 70% training data and 30% test data, with coverage targets of 50% clear daytime traffic conditions, 25% adverse weather conditions, and 25% high-traffic-density scenarios.

Closed-Loop Evaluation Metrics

Evaluator-free closed-loop requirements

Metric	Target
Correlation of internal Latent Closed-Loop score vs. external NAVSIM/CARLA-based score (Pearson r)	≥ 0.85
Sequential e-Testing time reduction vs. fixed-sample testing	40% – 60%
Type-I error (false-pass rate) under sequential testing	$\leq \alpha = 0.05$
Total GPU FLOPs reduction, full pipeline vs. Drive-JEPA + external simulator baseline	60% – 80%

Mandatory Demo Scenarios (Vietnamese Dense-Traffic Specific)

Mandatory demo scenarios

No.	Scenario Name	Description
1	Motorcycle side-cut avoidance	Motorcycle cutting in from blind spot at low speed

2	Dense market-street weaving	Narrow street, > 15 motorcycles in frame
3	Residential alley merge	Narrow alley merging into main road, no lane markings
4	Mixed-traffic intersection	4-way, no traffic light, negotiated right-of-way
5	Pedestrian + motorcycle mix	School-zone-like crossing with mixed traffic
6	Rain + dense traffic	Reduced visibility, adverse weather
7	Night residential driving	Low light, unmarked lanes
8	Sudden stop (lead vehicle)	Hard-braking lead motorcycle/car
9	Parked-vehicle door/obstacle	Static obstacle avoidance, narrow clearance
10	Full route, mixed conditions	10 – 15 min end-to-end residential + arterial route

Success Criteria

- 10/10 scenarios pass in Latent Closed-Loop Simulation
- 0 simulated collisions, 0 critical rule violations
- Trajectory + rollout video delivered for each of the 10 scenarios

Documentation

- **Technical report:** minimum 20 pages
- **Comparison study:** correlation study methodology, e-testing derivation

Milestone 4: Fine-Tuning, Optimization, KPI Validation & Deployment Package

Deadline: June 30th, 2027

Optimization Targets (Measurable Improvements)

Model optimization requirements

Optimization	Target	Measurement
Inference speedup (TensorRT)	$\geq 2\times$	vs. PyTorch baseline
FP16 accuracy drop	$\leq 1\%$	vs. FP32 baseline
INT8 accuracy drop	$\leq 3\%$	vs. FP32 baseline
Model size reduction	$\geq 40\%$	Quantization + pruning
Memory footprint	$\leq 40\text{GB}$	GPU RAM on Jetson AGX Orin

Final Resource-Efficiency Validation (Cumulative Contribution of the 3 Mathematical Tools)

Model optimization requirements

Component	Mechanism	Target Savings	Measured
Frozen ViT + Zero-Init Fusion	Backprop only on FlowFormer ++ branch	$\geq 70\%$ backprop cost	<i>(filled in at acceptance)</i>
Causal Factorization (do-operator)	z_{exo} computed once for 32 branches	30% – 40% rollout compute	<i>(filled in at acceptance)</i>
Anytime-Valid Rollout (Ville’s Inequality)	Early-exit high-risk branches	50% – 70% rollout compute	<i>(filled in at acceptance)</i>

Component	Mechanism	Target Savings	Measured
Sequential e-Testing	Real-time stopping on confidence	40% – 60% test time	<i>(filled in at acceptance)</i>

Deployment Deliverables

1. **TensorRT Engines:** camo_jepa_encoder.trt, scm_predictor.trt, latent_planner.trt
2. **Model Zoo (versioned, semantic versioning):**

```
models/
├── perception/
│   └── v1.0/{model.pth, model.onnx, model.trt, config.yaml}
```

 - scm_predictor/
 - latent_planner/
 - closed_loop_eval/
3. **Inference Benchmarks:** CSV — Model, Platform, Precision, Latency (ms), FPS, Memory (MB)
4. **Deployment Guide:** step-by-step Jetson AGX Orin installation, minimum 10 pages, troubleshooting section
5. **Docker Image:** $\leq 15\text{GB}$, tested on clean Ubuntu 22.04, includes ROS 2 Humble + TensorRT + all dependencies
6. **End-to-End Scenario Package** (per each of the 10 mandatory scenarios): trajectory JSON + rollout MP4 + risk-score log, delivered as a single archive

Evaluation Process

1. **Automated Tests:**
 - CI/CD pipeline
 - Unit tests $\geq 90\%$ pass, integration tests 95% pass, linting checks pass
2. **Benchmark Evaluation:**
 - Evaluate on the legacy dataset using NAVSIM and labeled data
 - Compare the results against Drive-JEPA baseline
 - Generate automated report

3. Latent Closed-Loop Simulation:

- Execute all 10 mandatory scenarios
- Log all metrics
- Generate success/failure report

Documentation

- **Technical report:** minimum 30 pages
- **Project coverage:** all 3 phases and full system architecture
- **Ablation study:** key component and design analysis
- **Resource validation:** resource-savings evaluation

DATA & HARDWARE REQUIREMENTS

Data Requirements

VinFast Must Provide

Data requirements from VinFast

Data Type	Minimum	Notes
Labeled frames (VF8)	1,000	Multi-sensor: cam + LiDAR + radar
New 8-camera multi-view raw video	$\geq 100\text{h}$	Dense residential + mixed-traffic routes, $\geq 70\%$ residential/market-street
CAN Bus / vehicle-state log	All sessions	Time-synced with video (steering, throttle/brake, speed, IMU)
HD maps	5 routes	OpenDRIVE format
Ground truth annotations	All frames	KITTI or nuScenes format

Delivery timeline: Week 1–2 of project start

Note on labeling: Because CaMo-JEPA is self-supervised, no new 3D bounding-box or HD-Map annotation is required for the new data — this is a

deliberate design choice to reduce VinFast’s data-preparation burden versus the earlier Drive-JEPA/NAVSIM-based pipeline.

Hardware Requirements

Development (HCMUT Provides)

HCMUT development hardware

Hardware	Specification
GPU workstation	2× RTX 4090 or 4× RTX 3090
CPU	AMD Ryzen 9 or Intel i9
RAM	≥ 64GB
Storage	≥ 4TB SSD
Network	10 Gbps Ethernet

Testing (VinFast Must Provide Access)

VinFast hardware access requirements

Hardware	Requirement
NVIDIA Jetson AGX Orin	For deployment testing, remote access or on-site
VF8 vehicle	Min 40 hours access for validation phase
Proving ground	Min 10 days access for real-world testing
CARLA server	Remote access or cloud instance
Network storage	Min 2TB for dataset sharing

Timeline:

- Week 1-2: Setup development environment
- Week 3-4: Data access and exploration
- Month 2-6: Continuous access for testing

- Month 8-10: Vehicle and proving ground access

RISK MITIGATION PLAN

Technical Risks

Risk assessment and mitigation

Risk	Level	Mitigation Strategy
New data volume arrives slower than 35h/month	High	Active-learning frame selection to maximize information from limited hours; transfer/pretrain flow branch on public driving video (e.g. BDD100K); heavy augmentation on the existing 20-min pilot for early iteration
Confounder/orthogonality not converging (MI leakage)	Medium	Adversarial MI minimization with hyperparameter search; fallback to simpler linear orthogonal projection; extended validation on synthetic confounder scenarios
SCM predictor drift over long rollout horizon	Medium /High	Shorter rollout horizon fallback (1.5s instead of 3s); ensemble predictors with uncertainty-aware early stop; conservative Ville's-inequality tuning during early milestones
Latent Closed-Loop score does not correlate well with external simulator	Medium	Recalibrate against reused legacy labeled set; blend with limited external CARLA sanity-check if correlation target is not met
GPU resource shortage for 8-cam + 32-branch training	Medium	Cloud GPU burst rental (A100) budgeted as optional line item; reduce rollout branch count (32→16) as fallback
Vehicle/hardware testing access delayed	Low	Prioritize Latent Closed-Loop validation; treat real-vehicle spot-check as optional, post-milestone validation

Risk	Level	Mitigation Strategy
Hardware unavailable	Low	Cloud GPU rental (AWS/Azure A100); simulation-only validation; budget for cloud: 100M VNĐ
Scope creep	Medium	Strict milestone gates; change-request process; weekly progress reviews

Contingency Plans

If Milestone 1-2 Fails Acceptance

- **Contingency buffer:** +2 weeks for revision
- **Alternative models:** minimum 2 backup factorization/predictor variants prepared
- **Minimum acceptable:** 90% of KPI targets (instead of 100%)
- **Additional support:** Consult with external experts

If VinFast Data Insufficient

- **Fallback:** Use public datasets exclusively
- **Legacy labeled set:** usable as-is for baseline comparison regardless of new-data volume
- **Data collection:** HCMUT can assist with 8-camera rig calibration and capture logistics

If Hardware Access Delayed

Quality Assurance

1. **Weekly Progress Reviews:**
 - Every Friday, 1-hour meeting
 - Review metrics, address blockers
2. **Monthly Milestone Reviews:**
 - Formal presentation to VinFast
 - Demo of current capabilities
 - Go/No-go decision for next phase

3. Code Reviews:

- All code reviewed by ≥ 2 team members
- Pull request process
- Automated CI checks

4. Testing Protocol:

- Unit tests: after each feature
- Integration tests: weekly
- End-to-end tests: bi-weekly

COST SUMMARY

Total Project Cost

Total project cost

Phase	Cost (VNĐ)
Phase 1 – Perception Scale-Up (3 months)	202,500,000
Phase 2 – Prediction & Planning (4 months)	318,000,000
Phase 3 – Closed-Loop Eval & Deployment (3 months)	202,500,000
Total (Labor)	723,000,000

Payment Schedule

Payment schedule

Payment	Milestone	Percentage	Condition
1	Contract signing	30%	Advance payment, kick-off
2	Milestone 1 complete	25%	Perception
3	Milestone 2 complete	30%	Perception & Confounder Pipeline

Payment	Milestone	Percentage	Condition
4	Milestone 3-4 complete	15%	Final optimization, handover
	Total	100%	

Payment amounts:

- Payment 1: 216,900,000 VNĐ (30%)
- Payment 2: 180,750,000 VNĐ (25%)
- Payment 3: 216,900,000 VNĐ (30%)
- Payment 4: 108,450,000 VNĐ (15%)

Additional Costs (Optional)

Cloud GPU rental (if needed):

- 4× A100 GPUs for 3 months: 70,000,000 VNĐ
- Included in proposal contingency

Hardware (if VinFast needs to purchase):

- Workstation (i9 + RTX 4090): $100,000,000 \text{ VNĐ} \times 2 = 200,000,000 \text{ VNĐ}$
- Jetson AGX Orin: $30,000,000 \text{ VNĐ} \times 2 = 60,000,000 \text{ VNĐ}$
- *Note: These are VinFast's investment, not included in project cost*

PROJECT TIMELINE

Project timeline summary

MS	Deliverable	Duration	Deadline (tentative)
0	Project kick-off	1 week	Sep 1, 2026
1	Perception Backbone at Scale	3 months	Nov 30, 2026
2	Latent SCM Predictor & Anytime-Valid Causal Planner	4 months	March 31, 2027

MS	Deliverable	Duration	Deadline (tentative)
3	Closed-Loop Eval & e-Testing	2 months	May 31st, 2027
4	Fine-Tuning, KPI & Deployment	1 months	June 30, 2027
	Total	10 months	

Gantt Chart (Simplified)

Project timeline (Gantt chart)

Task	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Kick-off	X									
Perception at Scale	X	X	X							
Latent SCM Predictor & Causal Planner				X	X	X	X			
Closed-Loop Eval								X	X	
Fine-Tuning & Deploy									(X)	X

Value Proposition

HCMUT offers:

- A working CaMo-JEPA prototype already implemented at project start, reducing Phase-1 risk
- A fully self-supervised pipeline requiring no new 3D/HD-Map labeling, directly cutting VinFast's data-preparation cost

- A causally-structured predictor that removes confounding bias present in naive behavior-cloning world models
- A simulator-free, labeling-free closed-loop evaluation methodology, reducing dependency on external NAVSIM/CARLA licensing and 3D annotation pipelines going forward
- Deep familiarity with Vietnamese urban/residential traffic patterns (motorcycle density, lane-less streets)

Success Metrics

Project success will be measured by:

1. **Successful integration:** Models run on VinFast hardware
2. **VinFast acceptance:** All milestones accepted by VinFast team
3. **On-time delivery:** All deadlines met
4. **On-budget:** No cost overruns
5. **End-to-end deliverable completeness:** trajectory JSON + rollout video delivered for all 10 mandatory scenarios

Next Steps

1. **VinFast reviews this proposal** (1 week)
2. **Clarification meeting** (1 meeting, 2 hours)
3. **Contract negotiation** (1 week)
4. **Contract signing** (by August 30, 2026)
5. **Project kick-off** (September 1, 2026)